



Vedansh Shrivastava

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B.Tech

Mechanical Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology Indore	7.43 (Current)	2023-Present
B.Tech. Minor	Indian Institute of Technology Indore	8.00 (Current)	2023-Present
Senior Secondary	CBSE Board	93.4%	2023
Secondary	CBSE Board	96.0%	2021

EXPERIENCE

- Vikram Sarabhai Space Centre (SPL), ISRO** Jan 2025 – Apr 2025
Data Analysis & Pipeline Engineering Intern Remote | Kerala, India | Certificate
 - Engineered and benchmarked 3 Radiative Transfer (RadTrans) data pipelines for Exoplanetary Atmospheres using JWST observational data, across **9+ structured comparison parameters**.
 - Built a comprehensive pipeline comparison & validation document; debugged multi-platform installation failures, **reducing researcher setup time by 30%** and eliminating recurring configuration errors.
 - Laid the data-analysis foundation for an ongoing research paper, applying structured ETL workflows to process and validate large-scale spectroscopic datasets.
- Drishti CPS Foundation** May 2025 – Jul 2025
Autonomous Systems & Data Pipeline Intern On-site | IIT Indore
 - Designed an end-to-end autonomous drone delivery system (30 km coverage) with real-time sensor data ingestion and on-flight error handling, **reducing cost & sample-to-test time by 95%**.
 - Built a data-driven GUI integrating live telemetry streams for monitoring and delivery control; delivered full working prototype within internship timeline.
- 13th Inter-IIT Tech Meet, IIT Bombay** Nov 2024 – Dec 2024
Contingent Member – ISRO Problem Statement On-site | Mumbai, India | Certificate
 - Processed and analysed large-scale satellite imagery datasets to achieve **sub-pixel resolution** using wavelet decomposition and X-ray classification for Lunar elemental mapping.

PROJECTS

- Uber Eats Delivery Analytics – Large-Scale Pipeline Engineering** Dec. 2025
Self-led Github
 - Designed a **Python-based ETL pipeline** on 45k+ records: applied Regex and Pandas for data cleaning across 20 variables, resolving NaN inconsistencies and improving data reliability to **97%**.
 - Engineered a geospatial computation pipeline isolating 4,359 unique delivery coordinates, **cutting runtime from 7 hours to 39 minutes (90% reduction)** via vectorised batch processing.
 - Segmented delivery performance across 22 Indian cities into 4 efficiency tiers; quantified impact of traffic density and adverse weather on latency using structured aggregation and statistical modelling.
- Spotify Music Data Analysis – EDA & Business Insight Pipeline** Dec. 2025
Self-led Github
 - Built a modular **EDA pipeline** on 4,351 tracks; correlated 10 audio features using Seaborn to establish a data-driven popularity model (75th percentile threshold).
 - Processed metadata for **2,000+ releases and 3,996 artists** to identify highest-growth genre segments (Electronic & Modern) for 2025; translated raw data into actionable business-level insights.

TECHNICAL SKILLS

- Languages & Tools:** Python (Pandas, NumPy, Matplotlib, Seaborn, PySpark), SQL, C++, Git, Jupyter, Power BI, Excel
- Data Engineering:** ETL/ELT Pipeline Design, Data Cleaning & Validation, Geospatial Pipelines, Batch Processing, Data Modelling
- Platforms & Concepts:** RDBMS, SQL, Delta Lake, FastAPI, RAG/FAISS, Medallion Architecture, Cloud
- AI / ML:** NLP, Predictive Modelling, Neural Networks, Decision Trees, Fine-tuning (QLoRA), FAISS Vector Search

KEY COURSES

- **Mathematics:** Linear Algebra, Calculus, Differential Equations, Complex Analysis, Numerical Methods
- **Computing & ML:** Machine Learning, Python, MLP, Neural Networks, Decision Trees, Linear Regression

POSITIONS OF RESPONSIBILITY

- **Co-Head**, Astronomy (AsSET) Club, IIT Indore *March 2025 – Present*
- **Head – Alumni Records & Management**, Alumni Cell, IIT Indore *March 2025 – Present*
- **Election Officer**, Election Commission, IIT Indore *2023–24*

ACHIEVEMENTS

- **Regional Finalist**, NXP-AIM Challenge, Top 100 selected among 600+ teams *2024*